

ESP32-P4 Sample Datasheet (T-008 fixture)

Table 1. MIPI-DSI Display Interface Pin Configuration

Pin	Signal	Direction	Default Level	Description
GPIO0	DSI_CLK_P	OUT	Low	MIPI-DSI clock lane positive
GPIO1	DSI_CLK_N	OUT	Low	MIPI-DSI clock lane negative
GPIO2	DSI_D0_P	OUT	Low	Data lane 0 positive
GPIO3	DSI_D0_N	OUT	Low	Data lane 0 negative
GPIO4	DSI_D1_P	OUT	Low	Data lane 1 positive
GPIO5	DSI_D1_N	OUT	Low	Data lane 1 negative
GPIO23	LCD_RESET	OUT	High	Panel hardware reset (active low)
GPIO24	LCD_BL_EN	OUT	Low	Backlight enable

Table 2. LCD Controller Register Map

Address	Register	Reset Value	Access	Function
0x00	CTRL_MODE	0x0000	R/W	Display controller mode select
0x04	H_RES	0x0400	R/W	Horizontal resolution (1024)
0x08	V_RES	0x0258	R/W	Vertical resolution (600)
0x0C	PIX_FMT	0x0002	R/W	Pixel format RGB565
0x10	FB_ADDR	0x00000000	R/W	Framebuffer base address
0x14	INT_STAT	0x0000	R	Interrupt status flags
0x18	INT_MASK	0xFFFF	R/W	Interrupt mask register

Table 3. Power Supply Requirements

Rail	Min (V)	Typ (V)	Max (V)	Current (mA)
VDD_CPU	0.95	1.00	1.05	500
VDD_IO	3.13	3.30	3.47	120
VDD_DSI	1.14	1.20	1.26	80
VDD_PLL	1.71	1.80	1.89	40

Table 4. Clock Source Configuration

Clock	Source	Frequency (MHz)	Divider	Target Peripheral
CPU_CLK	PLL0	400	1	HP CPU core
APB_CLK	PLL0	100	4	Peripheral bus
DSI_CLK	PLL1	500	1	MIPI-DSI PHY
SPI_CLK	APB	50	2	SPI master

Table 5. GPIO Alternate Function Mux

GPIO	AF0	AF1	AF2	AF3
GPIO30	UART0_TX	I2C0_SDA	PWM0	GPIO
GPIO31	UART0_RX	I2C0_SCL	PWM1	GPIO
GPIO32	SPI2_MOSI	SDIO_D0	PWM2	GPIO
GPIO33	SPI2_MISO	SDIO_D1	PWM3	GPIO
GPIO34	SPI2_CLK	SDIO_CLK	PWM4	GPIO